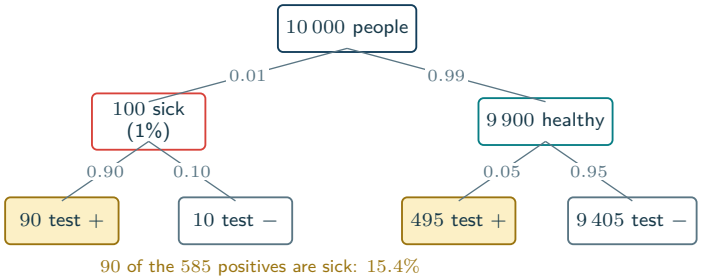
PROBABILITY LADDER 4 OF 9

Bayes' theorem and updating

Evidence turns a prior probability into a posterior one, and counting people in a table makes the arithmetic easy.

1 / THE RULE AND THE PICTURE

Most positives come from the large healthy group, so a positive test is far from conclusive



A disease affects 1% of people, the test catches 90% of the sick, and it wrongly flags 5% of the healthy.
Out of $10\,000$ people there are $90 + 495 = 585$ positives, so

$$P(\text{sick} \mid +) = \tfrac{90}{585} = 0.154$$

2 / BAYES' THEOREM / FROM THE MULTIPLICATION RULE

Since $P(A \text{ and } B) = P(A)P(B \mid A) = P(B)P(A \mid B)$, the two conditionals are related

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}, \qquad P(B) = P(B \mid A)P(A) + P(B \mid \text{not } A)P(\text{not } A).$$

Here $P(A) = 0.01$ is the prior, $P(B \mid A) = 0.90$ the sensitivity, and $P(B) = 0.9 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0585$, so the posterior is $\dfrac{0.9 \cdot 0.01}{0.0585} = 0.154$, exactly the table's $\frac{90}{585}$.

3 / THE TABLE OF COUNTS

Rows and columns give every conditional probability at once

	test positive	test negative	total
sick	90	10	100
healthy	495	9 405	9 900
total	585	9 415	10 000

Reading along the row, $P(+ \mid \text{sick}) = 90/100$, and reading down the column, $P(\text{sick} \mid +) = 90/585$; mixing up these two is the base-rate fallacy, and it happens because $P(+ \mid \text{healthy}) = 5\%$ is small while the healthy group is 99 times larger.

4 / ODDS FORM / REPEATED EVIDENCE

Posterior odds equal prior odds times the likelihood ratio

The prior odds are $1 : 99$ and the likelihood ratio is $\frac{0.90}{0.05} = 18$, so the posterior odds are $18 : 99 = 2 : 11$, a probability of $\frac{2}{13} = 0.154$.
A second independent positive test multiplies by 18 again, giving $36 : 11$ and a probability of $\frac{36}{47} = 0.766$, which is why repeated testing is far more decisive than a single result.

5 / THE MONTY HALL PROBLEM

Switching wins with probability $\frac{2}{3}$

You pick door 1 and the host, who knows where the car is, opens a goat door. Suppose he opens door 3: if the car is behind door 1 he chooses door 3 with probability $\frac{1}{2}$, giving weight $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$, while if it is behind door 2 he is forced to open door 3, giving weight $\frac{1}{3} \cdot 1 = \frac{1}{3}$, and behind door 3 he cannot.
The weights $\frac{1}{6} : \frac{1}{3}$ are $1 : 2$, so the car is behind your door with probability $\frac{1}{3}$ and behind door 2 with probability $\frac{2}{3}$. Counting the three equally likely car positions agrees: staying wins in one and switching in two.

ASIDE / PRIORS MATTER

A test with the same accuracy applied to a group where the disease affects 50% of people would make a positive result about 95% reliable.

LINKS TO OTHER SHEETS

The conditional rules are in *Conditional probability and independence*; the next sheet is *Random variables, expectation and variance*.

Convention. Sensitivity is $P(+ \mid \text{sick})$; the false positive rate is $P(+ \mid \text{healthy})$; the figures are illustrative.